To initialize a pointer, use the address-of operator (`&`):

```
int num = 10;

int *ptr = &num;  // ptr now holds the address of num
```

Dereferencing a Pointer

Dereferencing means accessing the value at the memory address the pointer holds. The dereference operator (`*`) is used for this purpose:

```
printf("%d", *ptr);  // Outputs 10, the value of num
```

Types of Pointers

- **Null Pointer:** Points to no valid memory location (usually defined as `NULL` or `0`).
- **Void Pointer:** A generic pointer that can point to any data type.
- **Wild Pointer:** An uninitialized pointer that points to some random memory location, leading to unpredictable behavior.

Dynamic Memory Allocation

Static memory allocation occurs at compile time, but dynamic memory allocation happens at runtime.